# Sharper One-Sample Matroid Prophet Inequalities

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**Written proof with same-agent verification; independent mathematical review and formalization pending.**

The refinement weights residual families by generator and witness counts, charges their entropy to expected sample size and nominal rank, and removes the density cutoff. The paper states the construction and source interfaces, proves the shared lemmas, and derives the selected coefficient below.

## 1 Current theorem and dependencies

For every known finite labeled matroid $M$ on $n$ elements, there is a measurable, distribution-independent online rule using one independent sample per label and $36n + 23$ initial independent fair bits. All sample/value coordinates are mutually independent and nonnegative, with matching laws at each label, and the seed is independent. Assume only that $\mathbb{E}[\mathrm{OPT}_M(V)] < \infty$. For every measurable arrival permutation depending on all samples, all values, and the entire seed, the accepted set is independent after every prefix and

$$\mathbb{E}[\text{reward}] \geq C_\star \mathbb{E}[\mathrm{OPT}_M(V)], \qquad C_\star = \frac{773895463001}{3203794364409053184000} > 2^{-32}.$$

The exact certificate is approximately **3.93235 times** the preceding `2^-34` certificate. This compares guaranteed fractions, not measured performance.

We retain the density paths, generators, hybrid safety comparison, marked-pivot machinery, rank accounting, and lossless simulation from OpenAI's pinned manuscript. Their interfaces are specified below. Computation and storage may be exponential, with exact observations; no polynomial-time claim is made.

## 2 The fixed-vector rule and its statistics

Fix a nonnegative hidden vector $w$ before any seed bits. The rule reveals and sacrifices a seed-chosen mask and must handle every permutation; its target is $\mathbb{E}_R \min_\pi w(A(w, R, \pi))$ . Write $\mathrm{r}(A \mid P) = \mathrm{r}(A \cup P) - \mathrm{r}(P)$ and $F_\varnothing = \mathrm{cl}(\varnothing)$ . Fix a label order for every tie. Higher rounded weight means higher priority. Put $W = \mathrm{OPT}_M(u)$ and let $I^*$ be priority greedy on the full rounded vector.

Let $\kappa \geq 4$ , $0 < h, \rho, t < 1$ , integer $L \geq 1$ , and $\theta > 1$ be fixed parameters supplied by the selected result. The independent masks $H, D, C, T$ have rates $h, \rho, t, t$ respectively. The main branch has probability $1 - t$ ; parity is fair. Choose a uniform shared phase $J \in \{0, \ldots, L-1\}$ , set $B = \theta^L$ , and define

$$u_J(x) = \theta^J B^{\lfloor \log_B(x/\theta^J) \rfloor} \quad (x > 0), \qquad u_J(0) = 0.$$

All random choices are made initially. Each result supplies exact dyadic mask probabilities and a finite fair-bit implementation. The score variables in the counting proof below are analysis devices, not algorithmic inputs.

The maximum branch reveals a separate fair mask and accepts the first unmasked positive nonloop with priority above every revealed positive nonloop. An empty threshold is below every positive nonloop. If $m_*$ is the largest rounded nonloop weight, expected worst-order reward is at least $m_*/4$ : the largest-priority nonloop is unmasked and the second is masked with probability $1/4$ . With only one positive nonloop success has probability $1/2$ ; with none the bound is zero.

In the main branch reveal and sacrifice $H \cup D \cup C$ . Define

$$U = \{e \notin H : u_e > 0,\ e \notin \mathrm{cl}(H_{>e})\}, \qquad Y = I^* \setminus H.$$

The candidate test uses only revealed $H$ values and the tested value. For a fixed phase divide weights by $\theta^J$ , so true groups $U_i$ have weight $B^i$ . List those meeting $D$ in increasing weight order as $G_1, \dots, G_m$ ; put $D_j = D \cap G_j$ , and similarly $C_j, T_j$ . The full groups are analysis objects; their revealed $D_j, C_j$ parts suffice for preparation. If there are no listed groups, reject all.

For a flat $P$ , let $Q_j(P)$ be the largest flat $Z \supseteq P$ maximizing $|D_j \cap Z| - \kappa\, \mathrm{r}(Z)$ . Take closure first for nonflat arguments. Set $a_j = -3j$ and enable $Q_{j,k} = Q_j$ for $k \geq a_j$ , with the identity earlier. Starting from $F_0(k) = F_\varnothing$ for $k < 0$ and $E$ otherwise, define

$$X_j(k) = Q_{j,k}(F_{j-1}(k)), \qquad F_j(k) = \mathrm{cl}(X_j(k) \cup (C_j \cap X_j(k+1))).$$

Compute paths in increasing $j$ on the finite relevant auxiliary timeline, and put $\widehat{F} = F_m$ . For a candidate nonloop in listed group $j$ , write $b_j(e) = \min\{k : e \in X_j(k)\}$ . It is eligible precisely when it is unsacrificed, $T$ -positive, has birth $b \geq a_j + 2$ of the chosen parity, and lies outside $\widehat{F}(b-2)$ . At endpoint $b$ maintain a set $A_b$ ; accept an eligible $e$ exactly when $\mathrm{r}(\{e\} \mid \widehat{F}(b-2) \cup A_b) = 1$ .

The source density and path lemmas give

$$F_{j-1}(k) \subseteq X_j(k) \subseteq F_j(k) \subseteq X_j(k+1), \qquad |D_j \cap (\mathrm{cl}(Q_j(P) \cup Z) \setminus Q_j(P))| \leq \kappa\, \mathrm{r}(Z \mid Q_j(P)).$$

Nonloops are born at $a_j$ or in $[a_j + 2, 0]$ , and the entire nominal path uses a generator set of size at most $|D_j|/\kappa$ . The nested layer bases give prefix independence. Finite rank operations, fixed tie rules, and countably many rounding levels give measurability. These structural facts are inherited from the cited source.

For valid selected-parity endpoints $a_j + 2 \leq b \leq 0$ , define

$$\begin{aligned}
K_{j,b} &= \bigcup_{\ell<j} (T_\ell \cap (X_\ell(b) \setminus X_\ell(b-1))),\\
P_{j,b} &= G_j \cap (X_j(b) \setminus X_j(b-1)), \quad O = D \cup C, \quad S_{j,b} = P_{j,b} \setminus O,\\
z_j &= \sum_b \mathrm{r}(S_{j,b} \mid X_j(b-2) \cup K_{j,b}),\\
\lambda_j &= \sum_b \mathrm{r}(T_j \cap S_{j,b} \mid \widehat{F}(b-2) \cup K_{j,b}).
\end{aligned}$$

The safe count $\sigma_j$ counts $d \in D_j \cap Y$ with such a birth and $d \notin Q_j(F_{j-1}(b-2) \cup K_{j,b})$ . For each nominal summand choose a fixed-order basis extension $Z_b$ , so $\sum_b |Z_b| = z_j$ . The residual is

$$\mathcal{R}_j = \bigcup_b (P_{j,b} \setminus \mathrm{cl}(X_j(b-2) \cup K_{j,b} \cup Z_b)) \subseteq O.$$

Extend $\sigma_i, z_i, \lambda_i$ by zero on unlisted true groups. The source density argument gives $\sigma_j \le \kappa z_j + |\mathcal{R}_j|$ . Its order-uniform lemma bounds the number of accepts of weight at least that of $G_j$ below by $\lambda_j$ , for every seed and order. One later-path generator budget over the whole timeline is

$$c_j = |C_j| + \sum_{\ell > j} (|C_\ell| + |D_\ell|/\kappa).$$

Conditional on $H, D, C$ , parity, and test bits outside the focal group, independent test thinning gives $\mathbb{E}[\lambda_j \mid \mathcal{F}_{-j}] \ge t(z_j - c_j)$ . These are the source interfaces used below.

## 3 A quadratic transaction inequality

**Quadratic transaction lemma.** Let $\mathcal{A}$ be an increasing event on finitely many independent bits, and let a predictable query procedure, reading each bit at most once, determine whether $\mathcal{A}$ occurs. Write $p = \mathbb{P}(\mathcal{A}^c)$ . At the completed transcript let $q$ be the failure probability with only unqueried bits random and every queried bit reset to zero. Then

$$\mathbb{E}[\mathbf{1}_{\mathcal{A}} q^2] \ge p(1-p).$$

**Proof.** At an intermediate transcript, let $W$ be its remaining coordinates and $I$ its queried positives. Let $m$ be the actual conditional failure probability, and $q$ the failure probability with only $W$ random and all other bits zero. Predictable product exposure identifies $m$ with the probability when $I$ is forced positive and $W$ is random. Increasingness gives $0 \le m \le q \le 1$ . Initially $m = q = p$ .

For a predictably chosen next coordinate, the depleted probability $q'$ is determined before the answer and satisfies $q' \ge q$ . The actual conditional failure probability is a martingale: $\mathbb{E}[m' \mid \text{transcript}] = m$ . Therefore the potential $\Phi(q, m) = q^2 - 2qm$ satisfies

$$\mathbb{E}[\Phi(q', m') \mid \text{transcript}] - \Phi(q, m) = (q' - q)(q' + q - 2m) \ge 0.$$

At the end $m = \mathbf{1}_{\mathcal{A}^c}$ , and failure forces $q = 1$ . The initial potential is $-p^2$ ; its final expectation is $\mathbb{E}[\mathbf{1}_{\mathcal{A}} q^2] - p$ . Sum the displayed query identity over the finite number of queries, padding stopped paths by zeros, to obtain the result. The proof includes $p = 0, 1$ and transactions needing no queries.

## 4 The reverse exit hazard

Fix the source's filter and density masks, a listed-group prefix, and a nonloop $d$ still alive before a reverse step. Its current states are $S$ , and its transition is $\mathcal{G}_k(c, S)$ . The remaining guard coordinates have their product law. Let $\mathcal{A}$ be the increasing event that $d$ survives, and let $p$ be the conditional exit probability.

For completeness, for $k < 0$ the source reverse map on a listed-group prefix is defined recursively by

$$A_0 = F_\varnothing, \qquad N_j = Q_{j,k}(A_{j-1}), \qquad A_j = \mathrm{cl}(N_j \cup \{e \in G_j \cap S_j : c_e = 1\}),$$

with output $(N_j)_j$ . Starting from $X_j(0) = E$ , reverse steps compute $X(k) = \mathcal{G}_k(C, X(k+1))$ . For actual later states, $N_j \subseteq S_j$ for every fresh bit assignment. After computing $N_j$ , query only $G_j \cap (S_j \setminus N_j)$ ; omitted additions already lie in $N_j$ . The new remaining domain is $\bigcup_j (G_j \cap N_j)$ .

The source queries exactly the departing coordinates in predictable batches. Their answers determine all output states: unqueried bits are inside flats already spanned. Thus the transcript decides $\mathcal{A}$ , on both survival and failure, so the quadratic transaction lemma applies.

Let $S'$ be the actual output and $V'$ the remaining coordinates. For each assignment $c$ supported on $V'$ , monotonicity in time and state gives

$$\mathcal{G}_{k-1}(c, S') \subseteq \mathcal{G}_k(c, S).$$

The right side has all just-queried coordinates zero, exactly the depleted experiment defining $q$ . Conditional on the full post-step transcript, the bits in $V'$ remain independent. On survival the next hazard $r$ therefore satisfies $r \geq q$ . The states are frozen in this comparison; earlier states are not recomputed after resetting bits. No order is conditioned on.

The next alive squared hazard consequently obeys

$$\mathbb{E}[\mathbf{1}_{\text{survive}} r^2] - p^2 \geq p - 2p^2.$$

Telescope this potential over the deterministic finite reverse horizon. Every nonloop exits, and if $P$ is its hazard at actual exit, then

$$-p_{\text{start}}^2 \geq \sum_i \mathbb{E}[\mathbf{1}_{\text{alive}} p_i] - 2 \sum_i \mathbb{E}[\mathbf{1}_{\text{alive}} p_i^2] = 1 - 2\mathbb{E}P.$$

Hence

$$\boxed{\mathbb{E}P \geq (1 + p_{\text{start}}^2)/2 \geq 1/2.}$$

In particular, the source's total squared-exit mass is at least $1/2$ . Its hybrid comparison still applies because guard and test rates agree. The same baseline-birth loss and fair parity give, after averaging a density mask of arbitrary rate $\rho$ ,

$$\mathbb{E}[\sigma_i \mid H] \geq \gamma |Y_i| - \frac{\rho n_i}{2\kappa}, \qquad \gamma = \rho/4.$$

This averages before conditioning on listing, extending the statistics by zero on unlisted groups.

The bound $1/2$ is sharp for the abstract hypotheses. Take $N$ Bernoulli-$t$ bits and repeatedly test the OR of the remaining suffix, querying until the first positive or until that suffix is exhausted. The next hazard equals the depleted current failure probability. With $a = 1 - t$ , classification by the last positive (or none) gives

$$\mathbb{E}P = a^{2N} + t \sum_{j=0}^{N-1} a^{2j} = a^{2N} + \frac{1 - a^{2N}}{2 - t}.$$

For $t = 1/m$ , $N = m^2$ , this tends to $1/2$ . This is an abstract limitation, not a hard matroid instance or a bound on the best prophet algorithm.

## 5 Joint residual encoding

Fix $H$ , parity, and all mask bits outside a true group $U$ of size $n \geq \kappa$ . Do not condition on its focal bits or listing. Its incoming path $F(k)$ , competition sets $K_b$ , index, and activation $a$ are fixed. Write $S_d(N) = \sum_{i=0}^{d} \binom{N}{i}$ . Suppose a path has density generators $J$ and rank witnesses $Z$ , of sizes $j, z$ . A generator has insertion time $\tau(g) \in [a, 0]$ , and

$$X(k) = \mathrm{cl}(F(k) \cup \{g \in J : \tau(g) \le k\}).$$

The source gives two marked streams. First, list $F(k)$ and then generators inserted at $k$ , marking the block one exactly when $k$ is a valid endpoint of the chosen parity. Second, start with the fixed $F(b_0)$ marked zero, where $b_0 \in \{a-2, a-1\}$ has that parity. At each subsequent parity endpoint $b$ , insert $K_b, Z_b$ with mark zero, then $F(b)$ and generators with $b-2 < \tau(g) \le b$ with mark one. End at $b_1 \in \{0,1\}$ of that parity; set $K_b = Z_b = \varnothing$ at invalid endpoints.

The second stream has intermediate span $\mathrm{cl}(X(b-2) \cup K_b \cup Z_b)$ and endpoint span $X(b)$ , since $K_b \subseteq F(b)$ and $Z_b \subseteq X(b)$ . The first test selects valid births. For a label passing it, the second pivot lies in the same layer and has mark one exactly on the residual. Thus the intersection of the two mark-one sets is the residual.

**Joint pivot lemma.** For fixed $J, Z$ , the number of residuals obtained from these linked streams is at most $3^j 4^{2j+z} S_{j+z}(2n)$ .

**Proof.** The source marked-pivot argument first bounds retained greedy bases of occurrences by $4^s$ when $s$ specified movable occurrences are inserted into an old sequence spanning the matroid. To recall why, old-only greedy gives an ordered basis. Only at most $s$ of its positions can be omitted after insertions: their conditional-rank drops for the full movable set sum to at most $s$ . There are at most $2^s$ choices of omitted old occurrences and $2^s$ of retained movable occurrences. Apply this to the two streams with $s = j$ and $s = j + z$ , giving $4^{2j+z}$ pairs of retained bases. Fix a pair. Each nonloop has a unique minimal support in each basis, by matroid submodularity, and its pivot is the last support member.

We represent both streams by the same $j + z$ variables. Label the entire ground set $0, \ldots, N-1$ , put $m = N + 1$ , $T = 8m$ , and $u_k = (k - a + 2)T$ . Give old $F(k)$ occurrences in the first stream absolute scores $2^{u_k+4m+e}$ , with sign equal to their fixed marks. In the second stream, the initial $F(b_0)$ has negative scores $-2^{u_{b_0}+4m+e}$ ; $K_b$ has scores $-2^{u_b+e}$ ; and $F(b)$ has scores $2^{u_b+4m+e}$ . These are fixed before the focal masks.

Give each generator a variable $x_g = 2^{u_{\tau(g)}+6m+g}$ . Its first score is $\eta_g x_g$ , positive exactly at a valid endpoint. Its second score is $d_g x_g$ , where $d_g = 1$ for the chosen parity and $d_g = 2^T$ otherwise. Multiplication by $2^T$ moves an insertion from $b-1$ into the endpoint $b$ block. Give each witness $e \in Z_b$ a variable $y_e = 2^{u_b+2m+e}$ , used only as the score $-y_e$ in the second stream. There are only three types for a generator:

$$(\eta_g, d_g) \in \{(1,1), (-1,1), (-1,2^T)\}.$$

The middle type includes early insertions of the chosen parity that are not valid births. Fix all types, at a cost of at most $3^j$ choices. Order every block by label. Second-stream generators can be reordered within their endpoint block because all have mark one; this preserves pivot marks and endpoint spans, and all retained bases are already counted.

In each stream the absolute scores are distinct powers of two in occurrence order. Every score dominates the sum of its predecessors. The sign of the sum over a support therefore equals its last member's mark. For fixed bases and types these are $2n$ affine functions of the same $j + z$ variables. Every realized mark pair is a strict sign pattern. The hyperplane-region bound gives at most $S_{j+z}(2n)$ patterns, proving the claim. Ground-set size and timeline length affect coefficients, not this dimension bound.

## 6 From a rank estimate to the final guarantee

The selected result must supply constants $K > 0$ and $\epsilon \geq 0$ such that, for each true group of size $n_i \geq \kappa$ ,

$$\mathbb{E}[z_i \mid H] \geq \frac{\gamma|Y_i| - \epsilon n_i}{K}, \qquad \gamma = \rho/4.$$

The source's later-path cost, including unlisted groups by domination, obeys

$$\mathbb{E}[c_i \mid H] \leq t n_i + (t + \rho/\kappa) \sum_{j>i} n_j.$$

Independent focal test thinning therefore gives

$$\mathbb{E}[\lambda_i \mid H] \geq \frac{t}{K} \left[ \gamma|Y_i| - (\epsilon + Kt) n_i - (Kt + K\rho/\kappa) \sum_{j>i} n_j \right].$$

Both changed cost factors are essential when $K \neq \kappa$ . For general filter rate $h$ , the source filter argument gives

$$\mathbb{E}_H u(Y) = (1-h)W, \qquad \mathbb{E}_H u(U) = \frac{1-h}{h} \mathbb{E}_H u(\text{greedy}(H)) \leq \frac{1-h}{h} W.$$

To verify the identity, condition on memberships preceding each label in rounded priority. Its span test is fixed. The candidate and sample-greedy probabilities are respectively $1-h$ and $h$ times this test; sum their weighted contributions. The comparison set $Y$ remains independent and contained in $U$ .

Set $\beta = (B-1)/B$ and

$$L_0 = \epsilon + Kt + \frac{Kt + K\rho/\kappa}{B-1}, \qquad s = (1-h)(\gamma - L_0/h).$$

For every fixed seed and arrival permutation, the source order-uniform rank bound implies that the number accepted at weight at least $B^i$ is at least $\lambda_i$ . An accepted weight $B^\ell$ contributes at most $\sum_{i \leq \ell} B^i = B^\ell/\beta$ to this cumulative sum. Consequently

$$\min_{\pi} u(A_{\text{main}}(R, \pi)) \geq \beta \sum_{i : n_i \geq \kappa} B^i \lambda_i.$$

The weighted higher-group cost is at most $u(U)/(B-1)$ . Discarding groups smaller than $\kappa$ loses comparison mass at most $\kappa m_*/\beta$ , where $m_*$ is the largest rounded nonloop weight. Applying the filter identities yields

$$\mathbb{E}_R \min_{\pi} \text{reward}_{\text{main}} \geq \frac{\beta t s}{K} W - \frac{t\gamma\kappa}{K} m_*.$$

If $(1-t)t\gamma\kappa/K \leq t/4$ , the maximum branch cancels the negative term. This condition is checked by the selected result's exact certificate. For positive $x$ , let $k = \lfloor \log_\theta x \rfloor$ . Across the $L$ phases its rounded values are $\theta^{k-r}$ , once each for $r = 0, \ldots, L-1$ . Since $\theta^k \geq x/\theta$ , the geometric sum gives

$$\mathbb{E}_J u_J(x) \geq \frac{\beta}{L(\theta - 1)} x.$$

Apply this to one fixed true optimal independent set to obtain the same factor for the rounded optimum. Thus, provided $s > 0$ , the coefficient is

$$C = \frac{(1-t)\beta^2 ts}{L(\theta-1)K}.$$

The minimum over permutations remains inside the seed expectation at every stage, including phase and branch. The source's glued-vector simulation reveals and sacrifices the mask fixed by the seed, using the sample there and the online value elsewhere. Conditional on the entire seed this glued vector has the original product value law, so it is independent of the seed. Ignored samples, sacrificed values, and unused coins may still inform the adversary's chosen permutation; the pointwise minimum covers them all.

# 7 Resources and proof dependencies

The rule uses a known finite labeled matroid, independent nonnegative values and matching independent samples, and exact observations. Only the expected offline optimum needs to be finite. Rank/closure tables and flat enumeration may be exponential; preprocessing and storage are part of the construction. There are finitely many occupied groups and a finite relevant auxiliary timeline. Observation precision and integer level indices are unbounded in the original measurable model. For rational inputs, exact power comparisons have costs depending on bit lengths and exponents. No polynomial complexity or practical performance claim is made.

The source density, generator, path, feasibility, hybrid safety, thinning, order-uniform rank, and simulation lemmas remain explicit dependencies. The quadratic and joint pivot proofs above are this project's written refinements. Exact certificates prove their scalar assertions, and finite checks test their specified cases. Independent mathematical review and formalization of these refinements remain pending.

# 8 Rank-charged residual entropy

This result retains the shared construction, quadratic hazard lemma, and joint pivot lemma in the framework. It replaces the density cutoff by a weighted count over both generator and witness sizes. The full one-sample and seed-visible adversary quantifiers are unchanged.

## 8.1 A weighted family bound

Use the framework's conditioning for a true group $U$ of size $n \geq \kappa$. For each fixed generator and witness pair $J, Z$ of sizes $j, z$, the joint pivot lemma gives at most $3^j 4^{2j+z} S_{j+z}(2n)$ residuals. That proof has no equal-cutoff hypothesis. Therefore there is a family fixed before focal masks for each $0 \leq j, z \leq n$, with

$$|\mathcal{F}_{j,z}| \leq N_{j,z} := \binom{n}{j}\binom{n}{z} 48^j 4^z S_{j+z}(2n).$$

Here $S_d(N) = \sum_{i=0}^{\min(d,N)} \binom{N}{i}$. All insertion times and witness-layer assignments are included in the joint affine count. Allowing arbitrary, possibly overlapping subsets only increases this bound. The actual witness union has size exactly the nominal rank $z$, because its layers are disjoint. Choose the source's generators by a fixed rule; their count satisfies $\kappa j \leq |D \cap U|$ on each listed outcome.

Let $p_O = \rho + (1-\rho)t$ and $\lambda = -\log p_O > 0$. Choose positive penalties $\mu, \nu$ and $0 < \xi, \eta, \zeta < 1$ satisfying

$$48e^{-\lambda\mu}/\zeta \le \xi, \qquad 4e^{-\lambda\nu}/\zeta \le \eta.$$

For every integer $d \ge 0$, including zero and $d \ge 2n$,

$$\zeta^d S_d(2n) \le (1+\zeta)^{2n}.$$

Indeed, $\zeta^d \le \zeta^i$ for $i \le d$; sum and extend the binomial expansion. Applying this at $d = j + z$ and summing over both sizes gives

$$\begin{aligned}\sum_{j,z=0}^{n} N_{j,z} e^{-\lambda(\mu j+\nu z)} &\le (1+\zeta)^{2n}\left(1+\frac{48e^{-\lambda\mu}}{\zeta}\right)^n\left(1+\frac{4e^{-\lambda\nu}}{\zeta}\right)^n \\ &\le (1+\xi)^n(1+\eta)^n(1+\zeta)^{2n} \le e^{nd_0}, \qquad d_0 = \xi+\eta+2\zeta.\end{aligned}$$

The last step uses $\log(1+x) \le x$. These are ordinary exponential weighting and binomial identities; the contribution is their use in this residual-to-rank estimate and the final theorem.

### 8.2 Adaptive residuals and both charges

On a listed outcome, let $R$ be the actual residual with actual counts $j, z$, and set $e = (|R| - \mu j - \nu z)_+$. Set $e = 0$ on unlisted outcomes. For real $x \ge 0$, the event $e > x$ implies that some fixed $R \in \mathcal{F}_{j,z}$ satisfies $R \subseteq O$ and $|R| > x + \mu j + \nu z$. For each fixed residual, the original conditional product law gives $\Pr(R \subseteq O) = p_O^{|R|}$. The union bound and weighted count yield

$$\Pr(e > x) \le \min\{1, e^{nd_0 - \lambda x}\}, \qquad \mathbb{E}e \le \frac{nd_0 + 1}{\lambda}.$$

The integral splits at $nd_0/\lambda$. The selected residual and its sizes may depend on the mask; no independence of that selection is asserted. The proof never multiplies this containment probability by a probability for the selected generator set, which could overlap the residual.

The source comparison and generator bound imply pointwise

$$\sigma \le \kappa z + |R| \le (\kappa+\nu)z + \mu j + e \le Kz + \frac{\mu}{\kappa}|D \cap U| + e, \qquad K = \kappa + \nu.$$

On unlisted outcomes $\sigma = z = e = 0$, so the final inequality also holds. Since $\mathbb{E}|D \cap U| = \rho n$, the framework's safety estimate gives its required rank interface with

$$\mathbb{E}[z_i \mid H] \ge \frac{\gamma|Y_i| - \epsilon n_i}{K}, \qquad \epsilon = \frac{\rho}{2\kappa} + \frac{\mu\rho}{\kappa} + \frac{d_0 + 1/\kappa}{\lambda}, \qquad \gamma = \rho/4.$$

Absorbing the additive one uses only $n_i \ge \kappa$. The inequality also holds for negative numerators by direct rearrangement. Both charges are necessary: the witness penalty increases $K$, and the generator penalty contributes $\mu\rho/\kappa$. There is no density cutoff or exceptional event.

### 8.3 Parameters and exact coefficient

Use

$$\kappa = 272, \quad \rho = 13/32, \quad h = 3/4, \quad t = 2^{-14}, \quad \theta = 261/256, \quad L = 256, \quad p_{\text{main}} = 1 - t,$$

and analysis-only constants

$$\mu = \nu = 39/2, \quad K = 583/2, \qquad \xi = 1/640, \quad \eta = 1/8192, \quad \zeta = 1/1280.$$

The exact certificate uses outward rational atanh-series logarithm enclosures after dyadic range reduction and verifies

$$\lambda > 9/10, \qquad \mu\lambda > \log \frac{48}{\xi\zeta}, \qquad \nu\lambda > \log \frac{4}{\eta\zeta}, \qquad B > 140.$$

The two penalty comparisons use the unrounded rigorous interval for $\lambda$. The coarser $9/10$ alone is insufficient for the witness comparison and is used only in the error below. With $d_0 = 133/40960$,

$$\bar{\epsilon} = \frac{\rho}{2\kappa} + \frac{\mu\rho}{\kappa} + \frac{d_0 + 1/\kappa}{9/10} = \frac{7847}{208896}.$$

Use $\beta > 139/140$, $B - 1 > 139$, and $L(\theta - 1) = 5$. The framework's slack satisfies

$$s > \frac{1}{4}\left[\frac{13}{128} - \frac{4}{3}\left(\bar{\epsilon} + Kt + \frac{Kt + K\rho/\kappa}{139}\right)\right] = \frac{1019519}{174219264} > 0.$$

The factors $Kt$ and $K\rho/\kappa$ account for the changed denominator. The small-group loss still uses the size threshold $\kappa$, giving $t\gamma\kappa/K$ times the maximum. Since $K \geq \kappa$ and $\gamma \leq 1/4$, the maximum branch cancels this loss. The final coefficient is strictly greater than

$$C_2 = \frac{773895463001}{3203794364409053184000} > \frac{33}{32}\,2^{-32}.$$

It is approximately $1.0374747334 \cdot 2^{-32}$, or 3.93234995 times the preceding exact certificate and 15.95344868 times the earlier $2^{-36}$ certificate. These compare guarantee coefficients, not measured performance.

The seed uses $2 + 5 + 14 + 14 + 1 = 36$ fair bits per label and $8 + 14 + 1 = 23$ shared bits. The density mask accepts 13 of 32 equally likely outcomes; filter, guard/test, and phase probabilities are likewise exact. Every coin is drawn initially, including unused-branch randomness. The analysis penalties and tilts require no algorithmic samples, bits, or computation.

### 8.4 Verification and review

The numerical search ran two attacks: charge witness entropy with the old cutoff, then charge generator entropy as well. It used 16 deterministic starts per mode and a declared 15-tuple grid. No global optimality claim follows from this search.

The finite checker covers 78 weighted generating bounds, 972 adaptive tail bounds, 216 integrated expectations, and 45,824 path charges on 170 matroid/weight instances. The path checks use small stress penalties and threshold 2 to exercise nonzero residuals and ranks. Two abstract negative controls show why omitting either charge is invalid; they are not hard matroid instances. The previous quadratic and joint-encoding checkers remain in the reproducibility suite.

The same-agent audit checks the stratified family, adaptive union bound, both charges, changed accounting factors, and full-seed simulation. Independent mathematical review and formalization are pending. Finite tests do not prove the universal theorem; its claim rests on the written proof and cited source interfaces. No exhaustive novelty or strongest-known-bound claim is made.

## 9 Sources, attribution, and comparison limits

OpenAI supplies the original qualitative constant guarantee and construction in *One Sample Suffices for Matroid Prophet Inequalities against an Almighty Adversary*, September 23, 2026, pinned at revision fd4aeeb2ee4f. Its density paths, reverse exposure, hybrid comparison, marked-pivot machinery, rank accounting, and lossless sample/value simulation are fundamental inputs. The manuscript derives $2^{-293}$, stronger than its $2^{-310}$ headline. The secretary consequence is in the same manuscript. Primary sources and hashes are recorded in the baseline.

This project's quadratic potential, joint affine encoding, and application of weighted residual counting refine that construction's quantitative analysis. Exponential weighting, union bounds, finite-offset rounding, and binomial estimates are established methods; no new general principle is claimed for them. Each version retains its specific evidence and source credit in the result registry and archives.

Daniel (@firsttimelifer) announced a $2^{-47}$ guarantee on October 8, 2026. A supporting manuscript was not located in the October 10 audit; its assumptions and proof remain unverified. It is acknowledged as an announcement, not an established comparison baseline. The 1/2 prophet result in Abdi, Banihashem, Hajiaghayi, and Mittal, arXiv:2609.19118v1 assumes a fixed order independent of samples and values and therefore uses a different adversary model.

This work was developed with Codex assistance. The paper's author is Ryan Shea. The written arguments and exact programs have same-agent verification; independent mathematical review and formalization of these refinements remain pending. No exhaustive priority search, confirmed best-known-bound claim, or practical performance claim is made. The repository uses Apache 2.0, with source attribution in NOTICE.

## 10 Development history

| Result | Guarantee | Change | Review stage |
|---|---|---|---|
| Parameter refinement | $> 2^{-63}$ | Refined source parameters, affine branch mixing, and finite rounding offsets. | Written proof; independent review pending |
| Logarithmic hazard potential | $> 2^{-43}$ | Introduced a logarithmic hazard potential and tighter residual accounting. | Written proof; independent review pending |
| Quadratic hazard potential | $> 2^{-36}$ | A quadratic exit potential, integrated residual tail, and revised masks and finite phases. | Written proof; independent review pending |

| Result | Guarantee | Change | Review stage |
|---|---|---|---|
| Joint residual encoding | $> 2^{-34}$ | Shared generator variables across two pivot tests and charged a density-sample cutoff. | Written proof; independent review pending |
| Rank-charged residual entropy | $> 2^{-32}$ | The refinement weights residual families by generator and witness counts, charges their entropy to expected sample size and nominal rank, and removes the density cutoff. | Written proof; independent review pending |

Earlier results retain their original evidence and source credit. These are stages of one research effort, not independent priority claims. Preserved publications are indexed in the archive.